

▼ **Figure 34.39** Short-beaked echidna (*Tachyglossus aculeatus*), an Australian monotreme. Monotremes have hair and produce milk, but they lack nipples. Monotremes are the only mammals that lay eggs (inset).



Marsupials

Opossums, kangaroos, and koalas are examples of the group called **marsupials**. Both marsupials and eutherians share derived characters not found among monotremes. They have higher metabolic rates and nipples that provide milk, and they give birth to live young. The embryo begins development inside the uterus of the female's reproductive tract. The lining of the uterus and the extraembryonic membranes that arise from the embryo form a **placenta**, a structure in which nutrients diffuse into the embryo from the mother's blood.

A marsupial is born very early in its development and completes its embryonic development while nursing (**Figure 34.40a**). In most species, the nursing young are held within a maternal pouch called a *marsupium*. A red kangaroo, for instance, is about the size of a honeybee at its birth, just 33 days after fertilization. Its back legs are merely buds, but its front legs are strong enough for it to crawl from the exit of its mother's reproductive tract to a pouch that opens to the front of her body, a journey that lasts a few minutes. In other species, the marsupium opens to the rear of the mother's body; in greater bilbies, this protects the young as their mother burrows in the dirt (**Figure 34.40b**).

Marsupials existed worldwide during the Mesozoic era, but today they are found only in the Australian region and in the Americas (**Figure 34.40c**). The biogeography of marsupials illustrates the interplay between biological and geologic evolution (see Concept 25.4). After the breakup of the supercontinent Pangaea, South America and Australia became island continents, and their marsupials diversified in isolation from the eutherians that began an adaptive radiation on the northern continents. Australia has not been in contact with another continent since early in the Cenozoic era, 66 million years ago. In Australia, convergent evolution has resulted in a diversity of marsupials that resemble eutherians in similar ecological roles in other

▼ **Figure 34.40** Marsupials in Australia and North America.



(a) A young brushtail possum (Australia). The offspring of marsupials are born very early in their development. They finish their growth while nursing from a nipple (in their mother's pouch in most species).



(b) A greater bilby (Australia). The greater bilby is a digger and burrower that eats termites and other insects, along with the seeds, roots, and bulbs of various plants. The female's rear-opening pouch helps protect the young from dirt as the mother digs. Other marsupials, such as kangaroos, have a pouch that opens to the front.



(c) Virginia opossum (Central and North America). This species is the only marsupial found north of Mexico. After the young complete development in the mother's pouch, they spend another four to five months riding on her back before living independently.